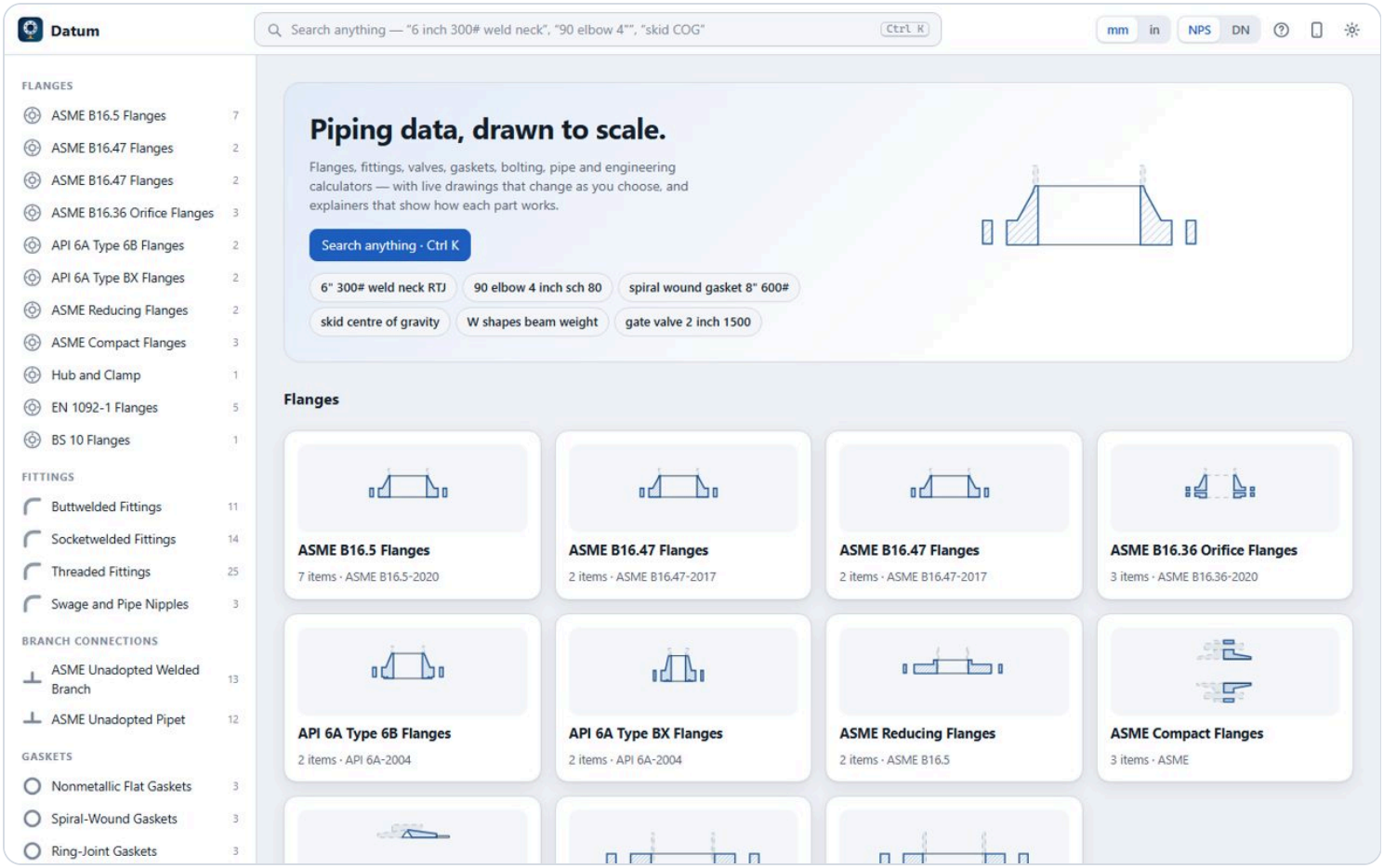
# Datum user guide

Datum is Dexterity Design Services' piping data book and engineering calculator. It runs on Windows and on Android phones, needs no network, and carries its whole data set inside the app.

This guide covers version 0.2 and later.



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# Install Datum

## Windows

1. Get `Datum-<version>-windows-x64-setup.exe` from the releases page (or the shared folder your team uses).
2. Run it. It installs for your user only and needs no administrator rights.
3. Start Datum from the Start menu.

The `.msi` file installs the same app; use it when IT deploys software centrally.

Datum is a single small program (about 4 MB) and uses very little memory. If Microsoft Edge WebView2 is missing (rare on Windows 10/11), the installer adds it.

## Android

There are two ways to get the Android app:

- **From a desktop on the office network:** open **Datum on a phone** in the desktop app (the phone icon at the top right) and scan the first QR code. The phone downloads `Datum.apk` from that PC. See Datum on a phone.
- **From the releases page:** download `Datum-<version>-android.apk` on the phone.

Open the downloaded file and allow "Install unknown apps" for your browser or file manager if Android asks. Datum needs Android 7.0 or newer. After installing, Datum works with no Wi-Fi or mobile data.

> Builds named `-unsigned-debug.apk` are test builds. They install normally, but a later build may need you to uninstall the old one first.

# Find what you need

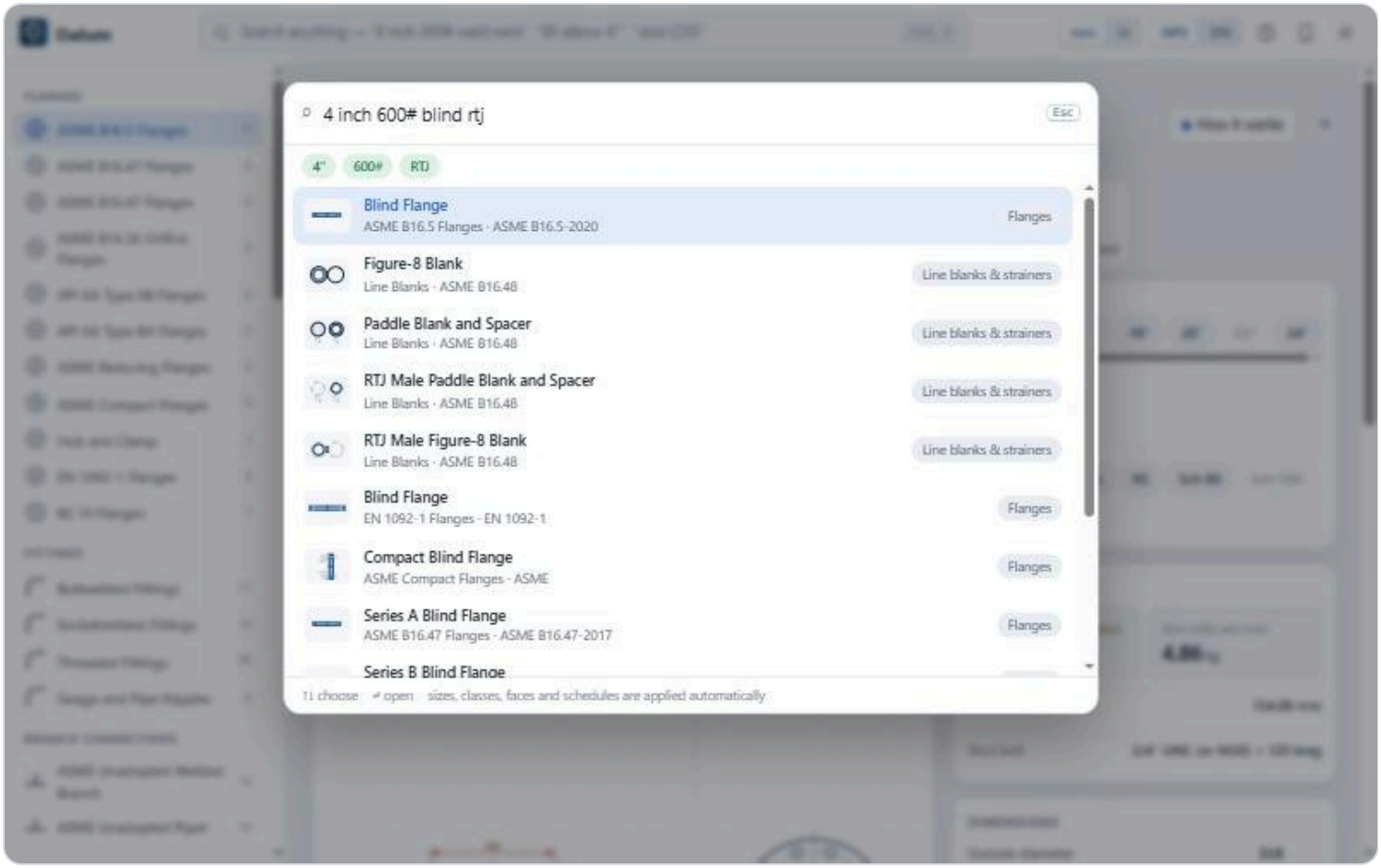
## Search anything

Click the search bar or press **Ctrl K** (or **/**). Type what you need in plain words:

| You type | Datum opens |
|---|---|
| `6" 300# weld neck RTJ` | ASME B16.5 weld neck flange, 6", Class 300, ring joint face |
| `90 elbow 4 inch sch 80` | Buttwelded 90° LR elbow, 4", Schedule 80 |
| `spiral wound gasket 8" 600#` | Spiral-wound gasket for B16.5, 8", Class 600 |
| `DN150 blind` | Blind flange (ASME B16.5) at DN 150 |
| `gate valve 2 inch 1500` | Gate valve, 2", Class 1500 |
| `skid centre of gravity` | Skid COG workbook |
| `W shapes beam weight` | Steel sections, W shapes |

Datum understands sizes ( `4"` , `4 inch` , `DN100` , `1-1/2` ), classes ( `300#` , `class 600` ), PN ratings, faces ( `RF` , `RTJ` , `FF` ), schedules ( `sch 40` , `STD` , `XS` ), angles and reducing sizes ( `6x4` ). The green chips under the search box show what it understood; those values are applied when the page opens. Common trade names work too, for example *spectacle blind*, *olet*, *swage*, *support span*.

Use ↑ ↓ to choose and **Enter** to open. **Esc** closes the search.




## Browse

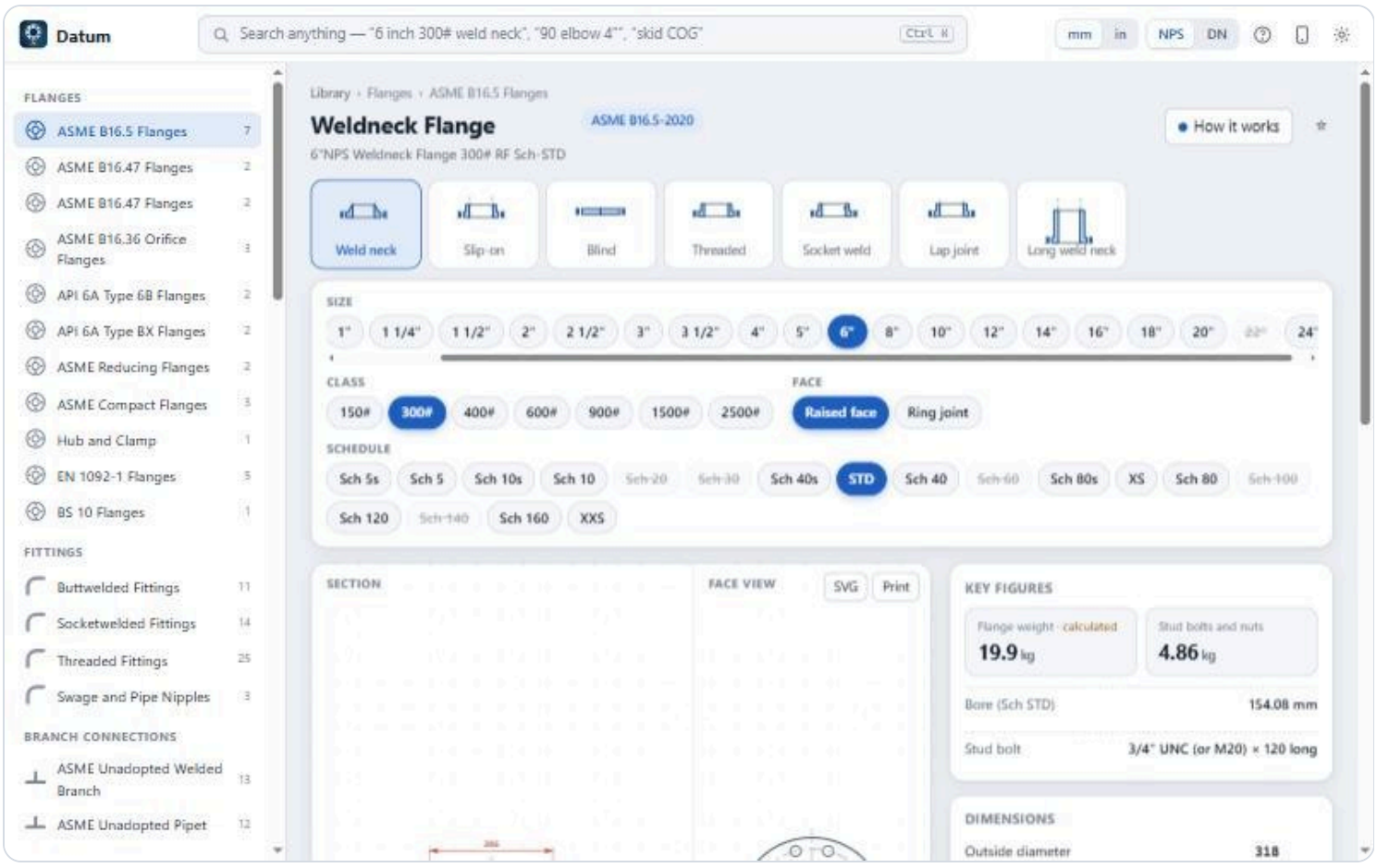
The left panel lists every family: flanges, fittings, branch connections, gaskets, valves, line blanks and strainers, pipe and bolting, piping engineering, and the two workbooks. The number next to each entry is how many component types it holds. On a phone, open the panel with the ☰ button.

The home page shows the same groups as cards, plus **Recently opened** and your **favourites**.

## Component pages

Each component page has, from top to bottom:

1. **Title and standard**: the component, its full description (for example *6" NPS Weldneck Flange 300# RF Sch-STD*) and the governing standard and edition.
2. **Type strip**: the other types in the same family (weld neck, slip-on, blind…). Click one to switch; the drawing changes shape to show the difference.
3. **Selections**: size, class, face, schedule and so on, as chips. Combinations that are not made in the standard are struck through and cannot be picked, so you never get an empty table.
4. **Drawing**: a section and a face view drawn to true scale from the selected row. It reshapes as you change the selection.
5. **Key figures**: calculated values such as weight and stud bolt length. Calculated values are labelled *calculated*.
6. **Dimensions**: every value from the standard for the selected size. Hover a dimension (or tap it on a phone) to highlight it on the drawing, and the other way round. Click ⧉ to copy a value.
7. **All sizes**: the full table for the current selections, with the selected size highlighted.




### Working with the drawing

- **Zoom:** mouse wheel or pinch. **Pan:** drag. **Fit:** double-click.
- On a phone the page scrolls normally until you pinch to zoom.
- Dimensions show in millimetres or inches, as set at the top right (**mm / in**).
- Labels keep the same readable size in every drawing, whatever the window or drawing size. Where labels would collide, Datum moves them clear of each other, of arrowheads and of drawn lines; a dimension too short for its arrows gets the arrows outside. Values are never changed.

### Favourites

Click ☆ next to the title to pin a page to the home page. Click ★ to unpin.

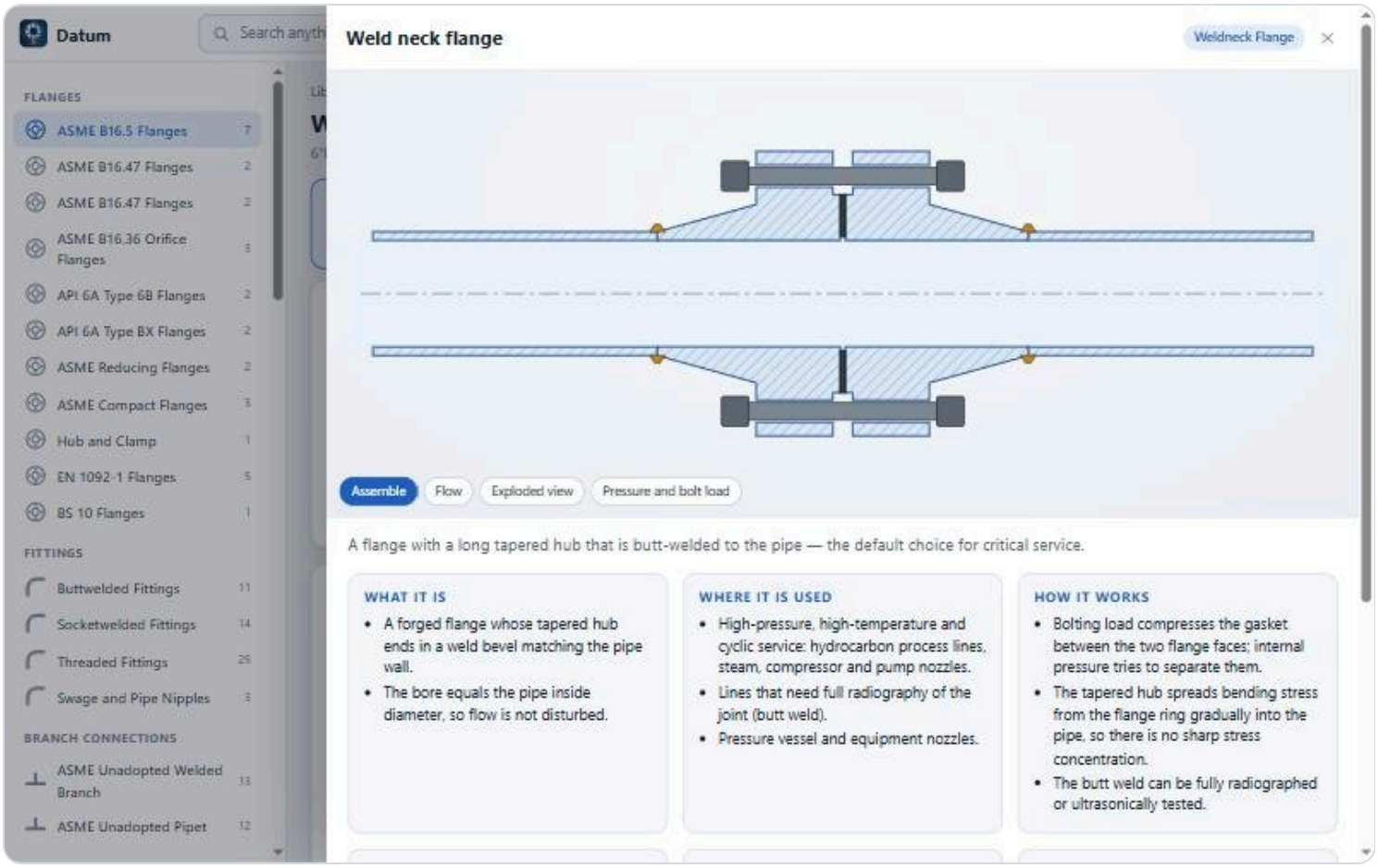
## How it works explainers

Click **How it works** on any component page. A large panel opens in the middle of the window:

- **Left:** the animated drawing of the part. The buttons under it switch between views such as *Open*, *Close*, *Throttle*, *Flow*, *Exploded view* and *Parts*, and the line under the buttons says what the current view shows.
- **Right:** the notes - **What it is**, **Where it is used**, **How it works**, **Benefits**, **Limits and cautions**, **Inspection and installation** - each in full. The chips at the top jump to a section; the notes scroll on their own, so the drawing stays in view.
- On a narrow window or a phone the notes follow below the drawing.

On a desktop, resting the mouse on a drawing for a moment shows a small preview card.

Close the panel with ×, **Esc**, or the phone's back gesture.

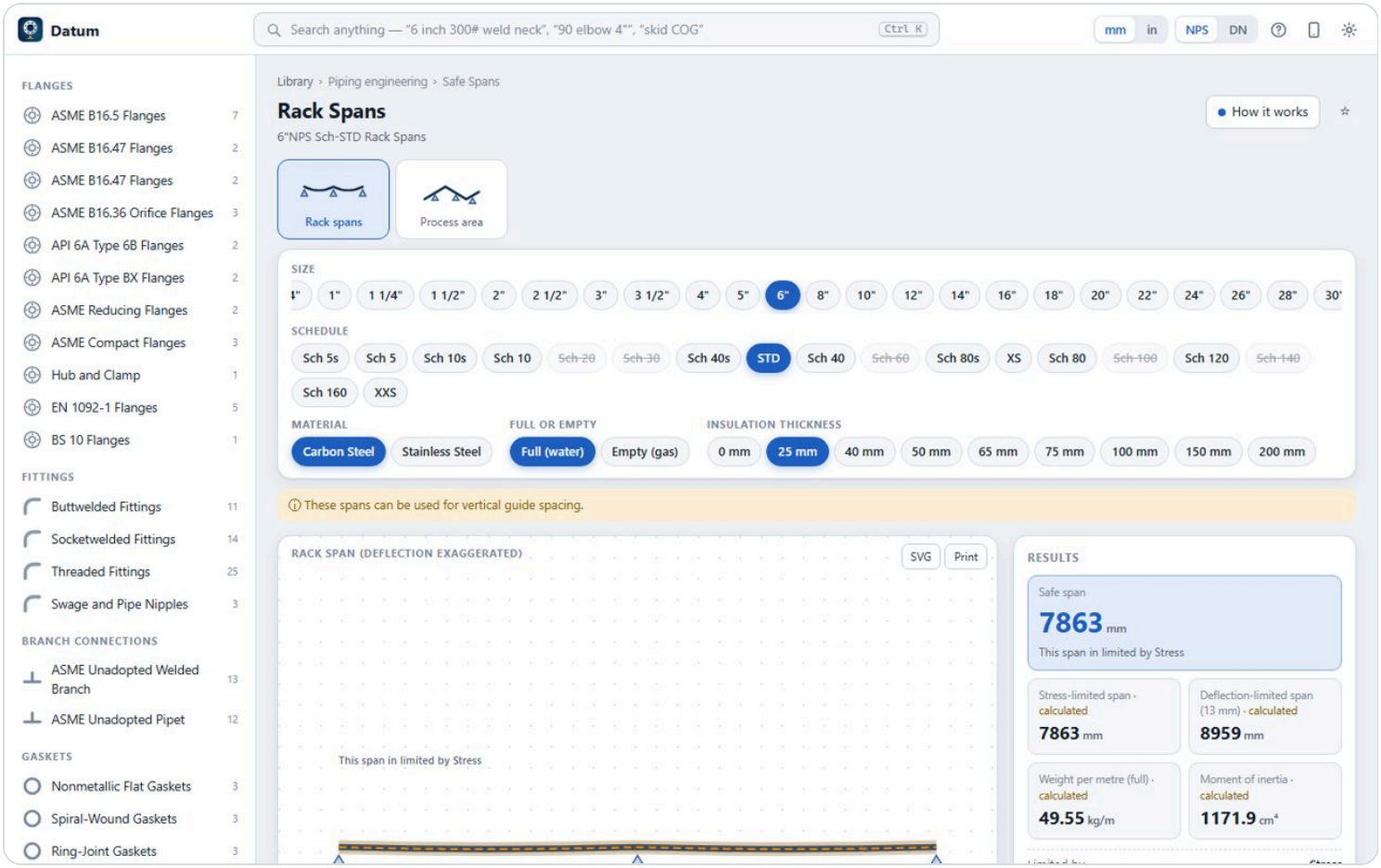


# Piping calculators

Open **Calculators** (home page or bottom bar on a phone):

| Calculator | What it does |
|---|---|
| Safe pipe spans | Support spans for pipe racks and process areas |
| Pipe flexibility | Leg length to the first guide, and expansion loop size |
| P-T ratings | ASME B16.5 flange pressure at temperature, by material group and class |
| Pipe spacing | Centre distances between lines, with or without flanges and insulation |
| ASME composite | Flange, gasket and bolting for one joint, together |
| Wall thickness | Pressure design wall and the schedule that satisfies it |
| Pipe size finder | Minimum line size for a flow at a velocity limit |

Each calculator has a live diagram. Inputs are on the left, results on the right; results change as you type.



## Structural & vessel workbook

This module is the office *Steel / Pipe / Vessel* workbook (37 sheets, including its hidden helper sheets) rebuilt as pages:

| Group | Pages |
|---|---|
| Steel sections | W, M, S, HP, channels, MC, angles, W tees, S tees, tubing |
| Bolting & grating | Structural bolts (grip and length, edge distance), welded grating |
| Weights calculator | Plate, pipe, bar, cone, angle and girder weights |
| Brace formulas | Vertical bracing lengths and work points (6 bay cases) |
| Triangle solvers | Right and oblique triangles, feet-inch converter |
| Feet-inch adding | Add and subtract feet-inch and millimetre lengths |
| Ladders, Stairs | Ladder, cage and rung layout; rise, tread, stringer and stair geometry |
| Vessel platforms | Circular platform bracket spacing on a vessel |
| Pipe tables, Pipe spans | Pipe weights, wall thickness, section properties; support spans |
| Elliptical / F&D heads | ASME 2:1 elliptical and flanged & dished head dimensions |
| Flanges by class | Run dimensions for classes 150 to 1500, RTJ, screwed fittings, weldolets, ELLs, tees, reducers, flange weights |

Calculator pages show the workbook's input cells (cell references in brackets, for example *H37*), a drawing, the results, and the hidden helper cells the workbook uses, so every number can be traced back to the sheet.

FLANGES
- ASME B16.5 Flanges 7
- ASME B16.47 Flanges 2
- ASME B16.47 Flanges 2
- ASME B16.36 Orifice Flanges 3
- API 6A Type 6B Flanges 2
- API 6A Type BX Flanges 2
- ASME Reducing Flanges 2
- ASME Compact Flanges 3
- Hub and Clamp 1
- EN 1092-1 Flanges 5
- BS 10 Flanges 1

FITTINGS
- Buttwelded Fittings 11
- Socketwelded Fittings 14
- Threaded Fittings 25
- Swage and Pipe Nipples 3

BRANCH CONNECTIONS
- ASME Unadopted Welded Branch 13
- ASME Unadopted Pipet 12

GASKETS
- Nonmetallic Flat Gaskets 3
- Spiral-Wound Gaskets 3
- Ring-Joint Gaskets 3



# Stairs

Sheet: Stairs

Stair design notes, stringer size, rise/tread table and WP-to-WP calculator

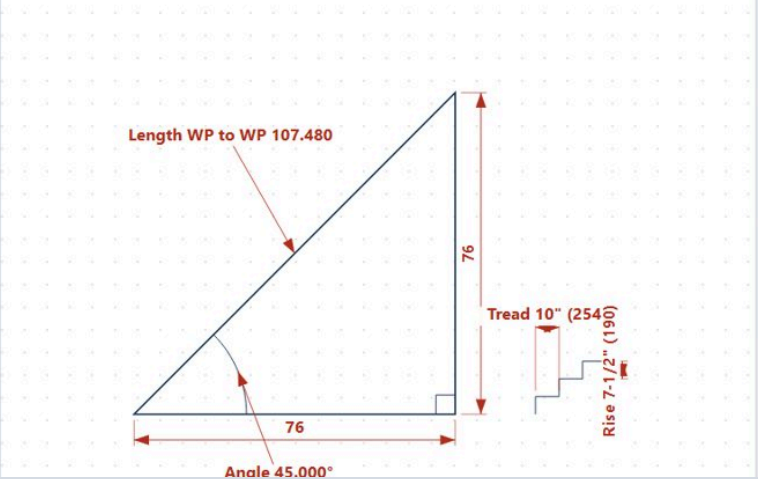
**Stair calculator (any consistent unit; result in the same unit)**

| HORIZONTAL RUN WP TO WP (H37) | VERTICAL RISE WP TO WP (J32) |
|---|---|
| 76 | 76 |

**Feet + Inches = Total in Feet**

| FEET (G24) | | INCHES (H24) | |
|---|---|---|---|
| 8 | ft | 6 | in |

STAIR - WORK POINT TO WORK POINT




**STAIR**

| Length WP to WP | Angle Degrees |
|---|---|
| 107.480 | 45.000 deg |

**CONVERTER**

| Total Feet | mm |
|---|---|
| 8.5000 ft | 2590.8 mm |

**HIDDEN HELPER CELLS O31:U32**

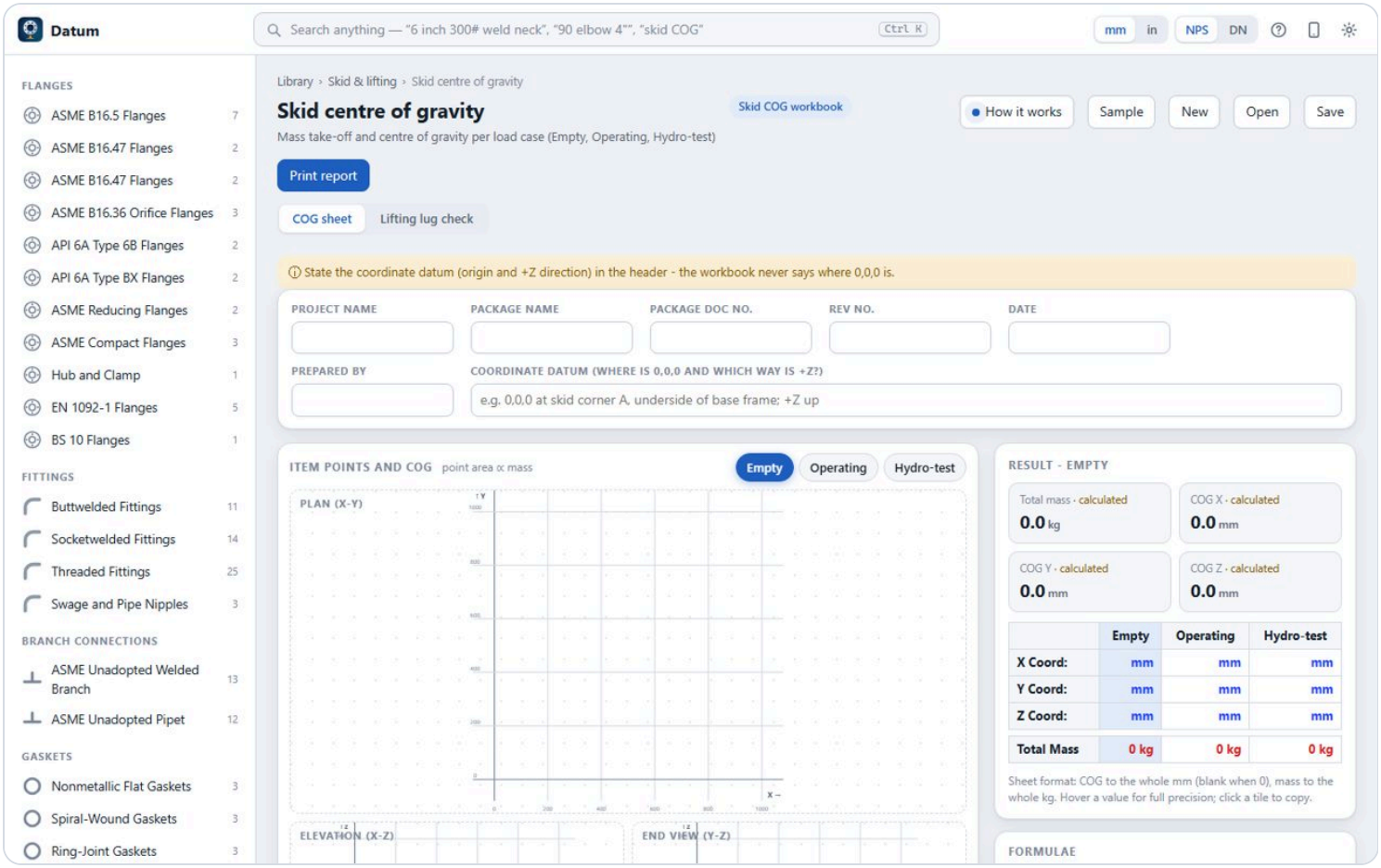
| b (O32) | c (P32) | a (R32) | A (S32) | B (T32) | C (U32) |
|---|---|---|---|---|---|
| 76 | 76 | 107.4802307 | 90 | 45 | 45 |

# Skid centre of gravity and lifting

This module is the *Skid COG calculation sheet* rebuilt.

## COG sheet

1. Fill in the header: project, package, document number, revision, date, prepared by, and the **coordinate datum** (where 0,0,0 is and which way +Z points). The workbook never stated this; Datum asks for it so the report is unambiguous.
2. Enter items: description, mass and X/Y/Z coordinates, with the load cases each item belongs to (**Empty**, **Operating**, **Hydro-test**).
   - **Paste rows…** takes a block copied from Excel (tab separated). You can also paste straight into any cell to fill from there.
   - **Import CSV / Export CSV** move the item list in and out.
   - Helpers fill items for pipe, flanges, tanks, segments and beams.
3. Results update as you type: total mass and COG for the chosen case, and a table of all three cases. The plan, elevation and end views plot each item with a dot sized by its mass, and the COG.
4. **Save** writes the job to a `.cog.json` file; **Open** loads one. **New** clears the sheet. **Sample** loads the workbook's own 251-item list.
5. **Print report** prints the sheet as a report.




## Lifting lug check

A four-point lift check: eye-pad loads from the COG, sling angle, and the lug and weld stresses, following the workbook's method.

## Settings

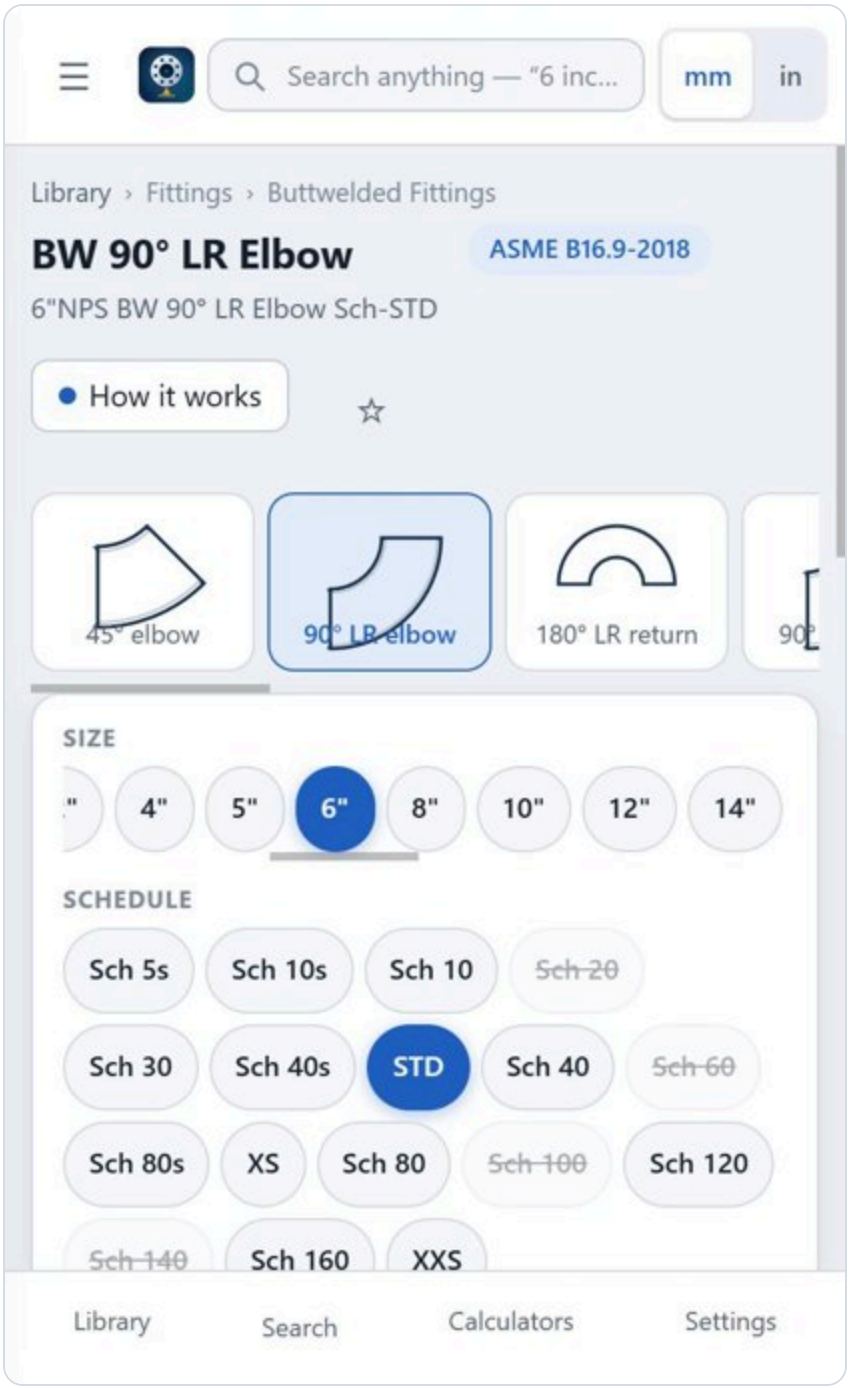
Open **Settings** (gear icon, or the bottom bar on a phone):

| Setting | Choices |
|---|---|
| Dimension units | Millimetres or inches (also the **mm / in** switch at the top) |
| Inch fractions | Decimal (1.5) or fractions (1 1/2) |
| Pipe size list | NPS (inches) or DN (millimetres) (also **NPS / DN** at the top) |
| Appearance | Auto (follows Windows or Android), Light, Dark |
| Data corrections | Shows whether the recorded source corrections are applied (see below) |
| Updates | Automatic or Manual checking, **Check now**, and the installed version (see Updates) |
| Reset | Clears selections, favourites and history on this device |

Settings are kept on each device separately.

## Datum on a phone

The Android app is the complete Datum: every table, drawing, calculator and workbook is stored in the app, and it works with no network.




On a phone:

- The bottom bar has **Library**, **Search**, **Calculators** and **Settings**; ☰ opens the full list.
- The **back gesture** (or back button) closes an open panel, search or menu first, then goes back a page. On the home page it leaves Datum.

- Saved files (CSV, drawings, COG jobs, this guide) go to **Downloads/Datum** (Android 10 and later; on older phones, to Datum's own folder under Android/data, shown when saving).

### Giving a phone the app from a desktop

In the desktop app, open **Datum on a phone**. Datum starts a small read-only server on the office network and shows two QR codes:

1. **Install the standalone app (Android):** scan, download `Datum.apk`, open it. This is the recommended way; after that the phone needs no network.
2. **Quick look in the phone browser:** any phone, including iPhone, while it is on office Wi-Fi and the PC is sharing. Nothing is installed.

Click **Stop sharing** when done. The first time sharing starts, Windows Firewall asks about Datum: allow it on **Private networks** only.

Sharing is read-only: phones can open the app and download the APK, nothing else. Nothing can be uploaded or changed on the PC, and Datum's data never changes over the network.

## Saving, exporting and printing

| Where | Button | Result |
|---|---|---|
| Component page | **Export CSV** | The all-sizes table as a CSV file for Excel |
| Component page | **SVG** | The drawing as an SVG file (opens in browsers, Inkscape, AutoCAD via import) |
| Component page | **Print** | A print-ready sheet with drawing and values |
| Workbook tables | **Export CSV** | The table as shown |
| Skid COG | **Save / Export CSV / Print report** | Job file, item list, report |
| User guide | **Download PDF** | This guide |

## Where the data comes from

Datum's tables and drawings come from the established piping data book the office used before, plus the two office workbooks (Steel / Pipe / Vessel, and the Skid COG sheet), converted in full, including their hidden sheets.

- **Datum reproduces the source exactly.** Where the source contains a known error (a typo in a table, a formula slip, a value that needs an engineering decision), Datum still shows the source value, and the problem is recorded in the errata register with the exact correction.
- The register currently holds 353 items, of which 112 await an engineering ruling. Settings shows whether any corrections are switched on.
- The data cannot be changed from inside the app or over the network. It changes only when a new Datum version is released.

If a value looks wrong, tell the Datum owner the page, the selections and the value; it can be checked against the register.

## Keyboard and touch shortcuts

| Action | Desktop | Phone |
| --- | --- | --- |
| Search | Ctrl K or / | Search in the bottom bar |
| Choose a search result | ↑ ↓ then Enter | Tap |
| Close a panel, search or dialog | Esc | Back gesture |
| Zoom a drawing | Mouse wheel | Pinch |
| Pan a drawing | Drag | Drag (when zoomed) |
| Fit a drawing | Double-click | Double-tap |
| Copy a value | Click ⧉ | Tap ⧉ |

## Updates

New versions of Datum are published automatically after every change. The Windows and Android apps look for them:

- a few seconds after Datum starts, then every 6 hours while it is open,
- when the PC wakes from sleep or comes back online,
- whenever you click **Check now** in Settings → Updates.

When a newer version exists, a bar appears under the search box:

- **Windows:** click **Install now**. Datum downloads the new version, checks its signature, closes, installs it (a small progress window, no questions) and opens again. Your settings, favourites and history are kept. **Later** hides the bar until Datum is next started.
- **Android:** click **Download**. The phone downloads the new APK; open it to install. Test builds (named `-unsigned-debug`) need the old app uninstalled first.

Set Settings → Updates to **Manual** to stop the automatic checks; **Check now** still works. Version 0.1.x has no updater: install 0.2.0 or later once by hand, and it keeps itself up to date. The browser copy (office Wi-Fi) always shows the version the sharing PC has.

## Troubleshooting

| Problem | What to do |
|---|---|
| Android will not install the APK | Allow "Install unknown apps" for the app you opened it from. If an older test build is installed, uninstall it first. |
| The phone cannot open the QR link | Phone and PC must be on the same office network; allow Datum in Windows Firewall on Private networks; check the PC is still sharing. |
| iPhone: Datum stops working away from the office | The iPhone browser view needs the office network. There is no iPhone app. |
| A size or class cannot be selected | It is struck through because the standard does not list that combination. |
| Values show in the wrong units | Use **mm / in** and **NPS / DN** at the top, or Settings. |
| The screen looks wrong after an update | Settings → Reset clears this device's selections and history. |
| "Could not check for updates" | The PC or phone is offline, or a proxy blocks raw.githubusercontent.com. Datum keeps working; try **Check now** later. |

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